

Econ 101A

Section 8

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1 Review: Slutsky Equation

When prices changes, the optimal consumption bundle changes. (We know this already.) The Slutsky equation decomposes that change into a **substitution effect** and an **income effect**.

1.1 Derivation

Consider the simple two-good case. We begin with the following equality between Marshallian ($x_i^*(p_1, p_2, M)$) and Hicksian ($h_i^*(p_1, p_2, \bar{u})$) demands, which, recall, is the dual condition.

$$x_i^*(p_1, p_2, e(p_1, p_2, \bar{u})) = h_i^*(p_1, p_2, \bar{u}) \quad (1)$$

We then differentiate both sides with respect to the price of good i .

$$\frac{\partial x_1^*(p_1, p_2, M)}{\partial p_1} + \frac{\partial x_1^*(p_1, p_2, M)}{\partial M} \times \underbrace{\frac{\partial e^*(p_1, p_2, \bar{u})}{\partial p_1}}_{\substack{=x_1^* \\ \text{(Envelope Theorem)}}} = \frac{\partial h_1^*(p_1, p_2, v(p_1, p_2, M))}{\partial p_1} \quad (2)$$

We then rearrange the equation and get the following.

$$\frac{\partial x_1^*(p_1, p_2, M)}{\partial p_1} = \underbrace{\frac{\partial h_1^*(p_1, p_2, v(p_1, p_2, M))}{\partial p_1}}_{\substack{\text{Substitution Effect} \\ <0}} - \underbrace{x_1^* \frac{\partial x_1^*(p_1, p_2, M)}{\partial M}}_{\substack{\text{Income Effect} \\ <0 \text{ if normal} \\ >0 \text{ if inferior}}} \quad (3)$$

This is the Slutsky equation. Notice that it provides a mathematical definition of the substitution effect and the income effect.

Using the same steps, we can also derive the more general Slutsky equation - i.e. where we have an arbitrary number of goods and where we differentiate *any* good with respect to *any* price.

$$\frac{\partial x_i^*(\mathbf{p}, M)}{\partial p_j} = \underbrace{\frac{\partial h_i^*(\mathbf{p}, v(\mathbf{p}, M))}{\partial p_j}}_{\text{Cross-substitution Effect}} - \underbrace{x_j^* \frac{\partial x_i^*(\mathbf{p}, M)}{\partial M}}_{\text{Cross-income Effect}} \quad (4)$$

Notice that now we can no longer definitively predict the sign of either the substitution effect or the income effect in the general scenario (if $n = 2$, the cross-substitution effect is positive).

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1.2 Terminology

- Two goods i and j are **gross substitutes** at price $\mathbf{p}$ and income M if

$$\frac{\partial x_i^*(\mathbf{p}, M)}{\partial p_j} > 0$$

- Two goods i and j are **gross complements** at price $\mathbf{p}$ and income M if

$$\frac{\partial x_i^*(\mathbf{p}, M)}{\partial p_j} < 0$$

- Warning:** Note that these do NOT need to be symmetric. That is, x_1 can be a gross substitute for x_2 and x_2 can be a gross complement for x_1 . For example:

$$\frac{\partial x_2^*(\mathbf{p}, M)}{\partial p_1} > 0 \text{ and } \frac{\partial x_1^*(\mathbf{p}, M)}{\partial p_2} < 0$$

- Two goods i and j are **net substitutes** at price $\mathbf{p}$ and income M if

$$\frac{\partial h_i^*(\mathbf{p}, v(\mathbf{p}, M))}{\partial p_j} > 0$$

- Two goods i and j are **net complements** at price $\mathbf{p}$ and income M if

$$\frac{\partial h_i^*(\mathbf{p}, v(\mathbf{p}, M))}{\partial p_j} < 0$$

- Unlike gross substitutes/complements, net complements/substitutes are *always* symmetric.¹

$$\frac{\partial h_i^*(\mathbf{p}, v(\mathbf{p}, M))}{\partial p_j} = \frac{\partial h_j^*(\mathbf{p}, v(\mathbf{p}, M))}{\partial p_i}$$

2 Labor Supply

The basic formulation of the labor-supply problem is as:

$$\max_{c, l} u(c, l) \text{ s.t. } pc \leq M + \underbrace{w(H - l)}_{=h}, \quad c \geq 0 \quad 0 \leq l \leq H$$

where:

- c is consumption.
- p is the per-unit price of consumption.
- H is the total number of hours in a day, week, month, whatever.
- l is leisure.
- h is hours spent working, which is just any hours not on leisure.
- w is the hourly wage.
- M is non-labor income.

¹You can prove this using the Envelope Theorem for the expenditure function.

Adding wl to both sides of the budget constraint, we can reformulate the question as:

$$\max_{c,l} u(c,l) \text{ s.t. } pc + wl \leq M + wH, \quad c \geq 0 \quad 0 \leq l \leq H$$

Notes:

- This shows that we can think that the labor-supply problem is just like all our other utility-maximization problems.
- Consumers optimize by choosing consumption and leisure.
- Leisure is just like any other good, which enters into the utility function and enters the budget constraint along with its implicit price (opportunity cost w).
- **Question:** We can apply the Slutsky equation to the labor supply question. What is the income effect and substitution effect of an increase in w on h^* (or $l^* = H - h^*$)?
- **Answer:** Substitution effect: work more (leisure less) when w increases because working pays more; Income effect: work less (leisure more) when w increases because it requires less work to reach a certain income.

3 Exercises

3.1 True or False?

1. A normal good can be a Giffen good.
2. Low-income people are more likely to take the bus than high-income people. That means that bus tickets are an inferior good.
3. If h^* is a Hicksian demand function, then $\partial h^*/\partial p$ isolates the income effect.
4. A normal good x is one such that $\partial x^*/\partial M > 0$, so that means that the income effect for a normal good is positive.
5. Bread bought and sold among peasants during the French Revolution is a plausible candidate for a Giffen good.

3.2 Slutsky Decomposition for Stone-Geary

1. Set up the expenditure-minimization problem for the utility function $u = x_2(x_1 + 4)$ and the constraint $u \geq U_0$.²
2. Drop the nonnegativity constraints and impose $u = U_0$. Set up the Lagrangian.
3. Find the FOCs.
4. Solve for h_1^* and h_2^* and assume an interior solution.
5. Find the Marshallian demand x_1^* and x_2^*
6. Are these goods normal or inferior?
7. Decompose the $\partial x_1^*/\partial p_1$ into a substitution effect and an income effect.
8. Now what about when $M < 4p_1$, in which case we have a corner solution? What are the Hicksian demands? What are the substitution and income effects?

²The most general formulation of Stone-Geary utility is $U(x, y) = (x - a)^\alpha (y - b)^\beta$. You saw Stone-Geary in PS3 and Section 6 (where are they?).

3.3 Slutsky Decomposition for Cobb-Douglas

Consider a consumer with Cobb-Douglas preferences:

$$u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$$

By assumption, $0 < \alpha < 1$. Recall that you already solved for the Marshallian demand functions (i.e., you already solved the *utility-maximization problem*):

$$x_1^*(p_1, p_2, M) = \frac{\alpha M}{p_1}, \quad x_2^*(p_1, p_2, M) = \frac{(1-\alpha)M}{p_2}$$

Similarly, you already solved for the Hicksian demand functions (i.e., you already solved the *expenditure-minimization problem*):

$$h_1^*(p_1, p_2, \bar{u}) = \bar{u} \left[\left(\frac{p_2}{p_1} \right) \left(\frac{\alpha}{1-\alpha} \right) \right]^{1-\alpha}, \quad h_2^*(p_1, p_2, \bar{u}) = \bar{u} \left[\left(\frac{p_1}{p_2} \right) \left(\frac{1-\alpha}{\alpha} \right) \right]^\alpha$$

1. Draw the **Engel curve** for x_1^* , meaning draw how x_1^* changes with M (holding prices constant).
2. Derive the indirect utility function $v(p_1, p_2, M) = u(x_1^*(p_1, p_2, M), x_2^*(p_1, p_2, M))$. Verify that $x_i^*(p_1, p_2, M) = h_i^*(p_1, p_2, v(p_1, p_2, M))$.
3. Derive the expenditure function $e(p_1, p_2, \bar{u}) = p_1 h_1^*(p_1, p_2, \bar{u}) + p_2 h_2^*(p_1, p_2, \bar{u})$. Verify that $x_i^*(p_1, p_2, e(p_1, p_2, \bar{u})) = h_i^*(p_1, p_2, \bar{u})$.
4. Decompose the $\partial x_1 / \partial p_1$ into a substitution effect and an income effect.
5. Verify that the Slutsky equation holds for $\partial x_1 / \partial p_1$. See equation 3 above.
6. Verify that the Slutsky equation holds for $\partial x_1 / \partial p_2$. See equation 4 above.